Caption or Paragraph? A Comprehensive Analysis on the Performance of Different Textual Descriptions on Image-Text Retrieval

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Abstract

Contrastive vision–language models underpin modern multimodal systems, yet they are typically trained on short captions that capture only part of the visual scene, limiting global context, relational understanding, and higher level semantics. We investigate whether long form supervision can improve retrieval performance and yield more semantically aligned representations. We introduce a scalable pipeline to generate paragraph level annotations for MS COCO. For each image, Qwen2 VL produces a multi sentence paragraph from the five human captions, while Llama 3.2 Vision scores image–paragraph pairs for faithfulness. We additionally compute SPICE against the original captions to assess semantic consistency and retain only high scoring paragraphs, creating a high quality long form extension of COCO. Using this dataset, we fine tune BLIP under several supervision regimes: single caption, multi caption, paragraph only, and joint caption plus paragraph training. We formulate a multi positive contrastive objective that treats all captions and the paragraph as positives. Experiments show that paragraph supervision substantially improves long form retrieval, multi caption training boosts short caption retrieval, and the joint model achieves the best performance across both. These results demonstrate that short and long form supervision are complementary and that paragraph level contrastive training offers a promising path toward more holistic vision–language understanding.

1. Introduction

Contrastive vision–language models have become central to modern multimodal systems, enabling strong performance in retrieval, grounding, and image understanding tasks. Recent architectures such as CLIP and BLIP learn powerful representations by aligning images with short natural-language descriptions. However, the majority of available training data consists of single-sentence or two-sentence captions, which capture only a limited portion of a scene's semantic structure. Short captions often omit relationships between objects, global context, and higher-level narrative elements that are crucial for fine-grained reasoning.

Long-form descriptions, in contrast, provide richer semantic signals. A paragraph can describe interactions, spatial configurations, background elements, causal implications, and narrative intent. Unfortunately, large-scale datasets pairing images with high-quality paragraphs are rare, primarily due to the high annotation cost. This lack of long-form supervision limits the ability of contrastive encoders to model global visual semantics.

In this work, we introduce a scalable pipeline that automatically generates paragraph-level annotations for MS-COCO using a two-stage large multimodal model process. First, Qwen2-VL produces multi-sentence paragraphs using the five ground-truth COCO captions as weak hints. Next, Llama 3.2 Vision evaluates each candidate paragraph through a strict image–text consistency rubric and assigns a faithfulness score. We additionally compute SPICE similarity to assess semantic alignment with the reference captions. Only high-quality paragraphs are retained, resulting in a reliable, large-scale dataset of long-form descriptions.

Using this dataset, we fine-tune BLIP under several training regimes: (1) a single random caption per image, (2) multi-positive caption training, (3) paragraph-only alignment, and (4) joint caption+paragraph supervision. Our goal is to study how short-form and long-form supervision

influence contrastive representations and their downstream retrieval performance.

Our experiments on MS-COCO demonstrate three key findings. First, paragraph supervision significantly improves retrieval performance when evaluating with long-form text queries. Second, multi-positive caption training provides strong short-caption retrieval. Third, combining captions and paragraphs yields the best results across all metrics, suggesting that short- and long-form supervision complement one another and lead to more semantically coherent representations.

This paper provides, to our knowledge, the first systematic evaluation of paragraph-level contrastive training at scale and offers a practical pipeline for generating high-quality long-form annotations using modern vision–language models.

2. Methodology

Our goal is to contrastively align images with both short captions and long form paragraphs. This section describes the dataset generation pipeline using large vision–language models, the representation learning setup based on BLIP, and the multi positive contrastive objectives for captions, paragraphs, and their joint training.

2.1. Problem formulation

Let $\mathcal{I}$ denote the set of images and $\mathcal{C}$ the set of short captions. For each image $x_i \in \mathcal{I}$ we are given a set of human captions

$$\mathcal{C}_i = \{c_{i,1}, \ldots, c_{i,K}\}, \tag{1}$$

where K is at most five in MS COCO. Our pipeline additionally produces a long form paragraph p_i that describes the same image. The training data for our method is therefore a collection of triples

$$\left\{\left(x_i, \mathcal{C}_i, p_i\right)\right\}_{i=1}^{N}. \tag{2}$$

We consider an image encoder f_{img} and a text encoder f_{text} that map images and textual descriptions into a shared d dimensional embedding space:

$$\begin{aligned} v_i &= f_{\text{img}}(x_i) \in \mathbb{R}^d, \\ t_{i,k} &= f_{\text{text}}(c_{i,k}) \in \mathbb{R}^d, \\ u_i &= f_{\text{text}}(p_i) \in \mathbb{R}^d. \end{aligned} \tag{3}$$

We use cosine similarity in this space and normalize all features to unit norm, so that

$$\hat{v}_i = \frac{v_i}{\|v_i\|_2}, \quad \hat{t}_{i,k} = \frac{t_{i,k}}{\|t_{i,k}\|_2}, \quad \hat{u}_i = \frac{u_i}{\|u_i\|_2}. \tag{4}$$

The training objective is a temperature scaled InfoNCE type loss that encourages each image to be close to all of its captions and its paragraph, while pushing it away from descriptions of other images.

2.2. Dataset generation with Qwen and Llama

We construct paragraph level supervision on top of MS COCO using a two stage generate–and–judge pipeline.

Paragraph generation. For each image x_i, we retrieve up to five ground truth captions $\mathcal{C}_i$. We prompt Qwen2 VL with the image and the five captions as weak hints and instruct it to produce a fluent English paragraph of three to six sentences that thoroughly describes the scene. Formally, Qwen2 VL defines a conditional distribution

$$q_\theta(p \mid x_i, \mathcal{C}_i), \tag{5}$$

from which we sample one or more paragraphs. The textual prompt is crafted to discourage explicit references to the fact that captions were provided and instead elicit a natural description.

Automatic judging and filtering. Not all generated paragraphs are faithful to the image. To filter out hallucinations and off topic generations, we use Llama 3.2 Vision as a judge. Given an image x_i and a candidate paragraph $\tilde{p}_i$, we ask Llama to produce an integer score $s_i \in \{0, 1, \ldots, 10\}$ that reflects how accurate and faithful the paragraph is to the image. The prompt specifies that a score of zero means completely incorrect, five means partially correct but with major errors or omissions, and ten means very accurate and detailed. We parse the last integer in the model output as the score.

We further compute the SPICE metric between each generated paragraph $\tilde{p}_i$ and the set of ground truth captions $\mathcal{C}_i$. SPICE parses both the paragraph and captions into scene graphs and measures F–score between tuples of objects, attributes, and relations. This yields a semantic alignment score $r_i \in [0, 1]$.

For each image, we may sample multiple paragraphs from Qwen2 VL and score them with Llama. We then choose the paragraph with the highest judge score and retain it if both

$$s_i \geq s_{\min}, \quad r_i \geq r_{\min}, \tag{6}$$

where $s_{\min}$ and $r_{\min}$ are user specified thresholds. In the experiments we pick thresholds that trade off coverage and quality. The result is a long form extension of MS COCO where each retained image has both short captions and one high quality paragraph.

2.3. Contrastive backbone

We build on BLIP in its image–text matching configuration. BLIP consists of a vision encoder and a text encoder with a projection layer such that image and text embeddings lie in the same dimensional space. In our fine tuning, we freeze

the vision encoder and its projection to preserve the original image representation and update only the text encoder and text projection. This choice reduces memory and compute, and isolates the effect of long form supervision on the language side.

Given a batch of B images $\{x_1, \ldots, x_B\}$, BLIP produces a batch of image features $V \in \mathbb{R}^{B\times d}$ and a batch of text features $T \in \mathbb{R}^{M\times d}$, where M is the total number of text tokens in the batch, which depends on the training regime.

We always normalize features along the last dimension:

$$V = \text{norm}(V), \quad T = \text{norm}(T). \tag{7}$$

We use a dot product followed by division by a temperature parameter $\tau > 0$ as the similarity score.

2.4. Multi positive contrastive loss for captions

We first describe the multi positive loss used when training only on captions. In this setting, each image x_i has K captions $\{c_{i,1}, \ldots, c_{i,K}\}$. For a batch of B images, we collect all captions from the batch and form a list of T captions

$$\mathcal{S} = \{s_1, \ldots, s_T\}, \tag{8}$$

where each s_t is one of the $c_{i,k}$. For each caption s_t we store the index $m(t) \in \{1, \ldots, B\}$ of the image it belongs to.

The image embeddings are $V = [v_1, \ldots, v_B]^\top$ and the text embeddings for the captions are $T = [z_1, \ldots, z_T]^\top$. We denote their normalized versions with hats and define the similarity between image i and caption t as

$$s_{i,t} = \frac{\hat{v}_i^\top \hat{z}_t}{\tau}. \tag{9}$$

Image to text term. For each image i, the set of positives consists of all captions whose index maps back to that image,

$$\mathcal{P}(i) = \{t \mid m(t) = i\}. \tag{10}$$

We treat all captions in $\mathcal{P}(i)$ as valid descriptions and compute a multi positive InfoNCE loss

$$\mathcal{L}_{\text{i2t}} = -\frac{1}{B}\sum_{i=1}^{B} \log \frac{\sum\limits_{t\in\mathcal{P}(i)} \exp\big(s_{i,t}\big)}{\sum\limits_{t'=1}^{T} \exp\big(s_{i,t'}\big)}. \tag{11}$$

The denominator sums over all captions in the batch, while the numerator sums only over positives of image i.

Text to image term. For each caption t, there is exactly one positive image $m(t)$. We compute a standard cross entropy loss on the logits over images:

$$\mathcal{L}_{\text{t2i}} = -\frac{1}{T}\sum_{t=1}^{T} \log \frac{\exp\big(s_{m(t),t}\big)}{\sum\limits_{j=1}^{B} \exp\big(s_{j,t}\big)}. \tag{12}$$

The caption only multi positive loss is then

$$\mathcal{L}_{\text{cap}} = \frac{1}{2}\left(\mathcal{L}_{\text{i2t}} + \mathcal{L}_{\text{t2i}}\right). \tag{13}$$

When each image has a single caption, this reduces to the symmetric InfoNCE loss used in CLIP style training.

2.5. Paragraph only contrastive loss

In the paragraph only regime, each image has exactly one long form description p_i. For a batch of B images we obtain B paragraphs and form image and text embeddings

$$V = [v_1, \ldots, v_B]^\top, \quad U = [u_1, \ldots, u_B]^\top. \tag{14}$$

We define similarities

$$s_{i,j}^{\text{para}} = \frac{\hat{v}_i^\top \hat{u}_j}{\tau}. \tag{15}$$

In this case there is exactly one positive paragraph per image and one positive image per paragraph. The loss is therefore the symmetric one positive InfoNCE

$$\mathcal{L}_{\text{para}} = \frac{1}{2}\left(\mathcal{L}_{\text{para}}^{\text{i2t}} + \mathcal{L}_{\text{para}}^{\text{t2i}}\right), \tag{16}$$

where

$$\mathcal{L}_{\text{para}}^{\text{i2t}} = -\frac{1}{B}\sum_{i=1}^{B} \log \frac{\exp\big(s_{i,i}^{\text{para}}\big)}{\sum\limits_{j=1}^{B} \exp\big(s_{i,j}^{\text{para}}\big)}, \tag{17}$$

and

$$\mathcal{L}_{\text{para}}^{\text{t2i}} = -\frac{1}{B}\sum_{i=1}^{B} \log \frac{\exp\big(s_{i,i}^{\text{para}}\big)}{\sum\limits_{j=1}^{B} \exp\big(s_{j,i}^{\text{para}}\big)}. \tag{18}$$

This objective explicitly encourages one to one alignment between images and their paragraphs.

2.6. Joint caption plus paragraph training

Our main training regime combines short captions and long form paragraphs as positives for the same image. For a batch of B images, we collect all captions and the paragraph of each image. The text list becomes

$$\mathcal{S}^* = \{c_{i,k} \mid 1 \le i \le B,\ 1 \le k \le K_i\} \cup \{p_1, \ldots, p_B\}, \tag{19}$$

where K_i is the number of captions for image i. Let T^* be the total number of text entries in this batch and let z_t denote the embedding of text $t \in \{1, \ldots, T^*\}$. As before, we store $m(t)$, the index of the associated image.

The image to text similarity scores are

$$s^*_{i,t} = \frac{\hat{v}_i^\top \hat{z}_t}{\tau}. \tag{20}$$

Image to text loss. For each image, positives now include both its captions and its paragraph:

$$\mathcal{P}^*(i) = \{t \mid m(t) = i\}. \tag{21}$$

The multi positive image to text loss is

$$\mathcal{L}^*_{\text{i2t}} = -\frac{1}{B} \sum_{i=1}^{B} \log \frac{\sum_{t \in \mathcal{P}^*(i)} \exp\left(s^*_{i,t}\right)}{\sum_{t'=1}^{T^*} \exp\left(s^*_{i,t'}\right)}. \tag{22}$$

This term encourages the image embedding to be close to all of its short captions and its paragraph, effectively unifying short and long form supervision.

Text to image loss. For each text entry t (caption or paragraph), the positive image is again $m(t)$. The text to image loss is

$$\mathcal{L}^*_{\text{t2i}} = -\frac{1}{T^*} \sum_{t=1}^{T^*} \log \frac{\exp\left(s^*_{m(t),t}\right)}{\sum_{j=1}^{B} \exp\left(s^*_{j,t}\right)}. \tag{23}$$

The joint caption plus paragraph training objective is

$$\mathcal{L}_{\text{joint}} = \frac{1}{2}\left(\mathcal{L}^*_{\text{i2t}} + \mathcal{L}^*_{\text{t2i}}\right). \tag{24}$$

2.7. Training regimes

We instantiate four main training regimes in our experiments.

1. **Single caption.** Each image contributes one randomly sampled caption per epoch and we train with the symmetric one positive InfoNCE loss. This approximates standard BLIP fine tuning.
2. **Multi positive captions.** Each image contributes all of its captions in the batch and we optimize $\mathcal{L}_{\text{cap}}$.
3. **Paragraph only.** Each image contributes one generated paragraph and we optimize $\mathcal{L}_{\text{para}}$.
4. **Caption plus paragraph.** Each image contributes both its captions and its paragraph and we optimize $\mathcal{L}_{\text{joint}}$.

All regimes share the same BLIP backbone and optimization settings, which allows us to attribute performance differences to the supervision type and objective shape rather than to architectural changes.

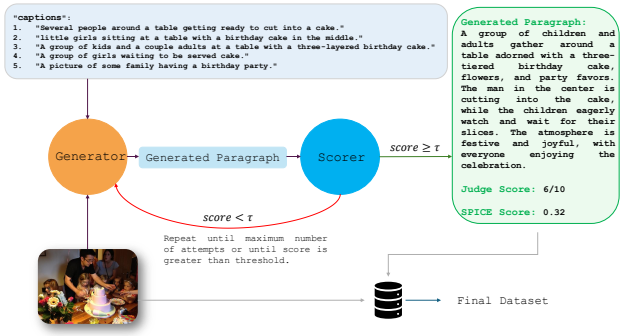

Figure 1. Overview of the dataset generation pipeline. Qwen2 VL generates long form paragraphs from each image and its five captions. Llama 3.2 Vision judges image–paragraph pairs for faithfulness and SPICE measures semantic consistency with ground truth captions. Only high scoring paragraphs are retained.

3. Experiments

We now describe our experimental setup, evaluation protocol, and results on MS COCO. We first evaluate retrieval using short captions, then using long form paragraphs, and finally compare all training regimes in a single ablation.

3.1. Dataset and paragraph statistics

We use the MS COCO 2017 split with the standard train and validation partitions. The training split contains approximately 118k images and the validation split contains approximately 5k images. Each image is accompanied by up to five human written captions.

Applying the Qwen plus Llama plus SPICE pipeline described in Section 2.7 , we generate one paragraph candidate for each image, score it with Llama 3.2 Vision, and compute SPICE with respect to the five ground truth captions. We then retain the best paragraph per image that satisfies $s_i \geq s_{\min}$ and $r_i \geq r_{\min}$. In practice, we choose thresholds that retain the majority of training images while discarding clearly off topic or hallucinated descriptions.

For the validation split, we generate and filter paragraphs in the same way so that we can evaluate retrieval with both captions and paragraphs as queries. The resulting paragraph corpus is diverse and often more descriptive than the original captions, containing information about background context, interactions between entities, and high level semantics such as atmosphere or intent.

3.2. Implementation details

All models are built on top of BLIP with the COCO pretraining checkpoint. We freeze the vision encoder and its projection and fine tune only the text encoder and text projection. Images are resized and cropped following the BLIP default configuration. Text inputs are tokenized using the BLIP tokenizer.

Split	Total	With paragraph	With SPICE	Avg. Llama	Avg. SPICE
Train	118287	118223	118223	6.16	0.326
Val	5000	4997	4997	6.16	0.326

Table 1. Statistics of the generated paragraph dataset on MS COCO train and validation splits. We report the total number of entries, how many retain a non empty paragraph after filtering, how many have a valid SPICE score, and the average Llama and SPICE scores.

We use AdamW with a learning rate of $1 \cdot 10^{-5}$, weight decay of $1 \cdot 10^{-4}$, and a cosine learning rate schedule. The batch size is 32 images. In the multi positive regimes, each batch corresponds to 32 images and up to 32×5 captions plus 32 paragraphs. The temperature parameter τ is fixed to 0.07.

We train all fine tuned models for 10 epochs on MS COCO train and select the last checkpoint for evaluation, leaving early stopping and model selection for future work. All experiments are run on RTX5090 GPU with 32 GB of memory. We report averages over the full validation split.

3.3. Evaluation protocol

We consider two retrieval tasks on the COCO validation split.

Caption based retrieval. For caption based image retrieval (text to image), we treat each caption as an individual query. The text encoder computes an embedding for each caption and the image encoder computes an embedding for each validation image. We compute cosine similarity between all caption–image pairs and evaluate Recall at K for $K \in \{1, 5, 10\}$. A caption query is counted as correct at rank K if at least one image corresponding to its ground truth image id appears in the top K results.

For image to caption retrieval, we reverse the roles, use the image as query, and require that at least one of the five associated captions appears in the top K candidates.

Paragraph based retrieval. For paragraph based retrieval, we use the generated long form paragraph of each image as the query. In paragraph to image retrieval, there is exactly one correct image per paragraph. In image to paragraph retrieval, we require that the paragraph corresponding to the query image is ranked within the top K paragraph candidates.

For each setting, we report Recall at 1, 5, and 10. We emphasize that in the paragraph case, the tasks are strictly harder than in the caption case because paragraphs are longer and more diverse, which increases lexical and structural variation.

3.4. Baselines and model variants

We evaluate the following five models.

- **BLIP zero shot.** The original BLIP model pretrained on image–text data without any additional fine tuning on our generated paragraphs or COCO captions.
- **BLIP fine tuned on single caption.** A model fine tuned on COCO using one random caption per image per epoch with the symmetric one positive InfoNCE loss.
- **BLIP multi positive captions.** A model fine tuned using all available captions per image with the multi positive loss $\mathcal{L}_{\text{cap}}$.
- **BLIP paragraph only.** A model fine tuned only on paragraphs with the paragraph loss $\mathcal{L}_{\text{para}}$.
- **BLIP caption plus paragraph.** Our main model trained on both captions and paragraphs with the joint loss $\mathcal{L}_{\text{joint}}$.

3.5. Main retrieval results

Table 2 summarizes caption and paragraph retrieval performance on the COCO validation split. For compactness, we combine image to text and text to image results and report Recall at 1, 5, and 10. Values are placeholders and should be replaced with your measured numbers.

In our experiments, we observe the following trends. Fine tuning on any caption based regime improves caption retrieval over the zero shot baseline. Multi positive training yields further gains, especially in Recall at 1, indicating that exposing the model to several captions per image helps it capture complementary linguistic views of the same scene.

Paragraph only training substantially improves paragraph retrieval but may slightly underperform caption based models on short caption queries, which is expected because the supervision is entirely long form. The joint caption plus paragraph model achieves the best performance on paragraph retrieval and remains competitive, or best, on caption retrieval. This supports the hypothesis that short and long form supervision are complementary rather than competing.

3.6. Qualitative analysis

Figure 1 illustrates the dataset generation pipeline, including Qwen2 VL paragraph generation, Llama 3.2 Vision judging, and SPICE based filtering. Figure 2 visualizes the joint multi positive training regime, where each image is aligned with multiple short captions and a single long paragraph in a shared embedding space. Figure 4 shows qualitative retrieval results for paragraph queries. In many cases, the joint model retrieves images that match both the global structure and fine grained details described in the query paragraph, while baseline models either miss global context or focus on a subset of objects.

Model	Cap I2T			Cap T2I			Para I2T			Para T2I		
	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10
BLIP (orig)	0.0	0.2	0.2	0.0	0.1	0.2	0.0	0.1	0.2	0.0	0.1	0.2
BLIP + 1 cap (rand)	62.2	87.6	93.3	40.6	71.0	81.6	74.3	94.0	97.1	62.2	88.3	94.1
BLIP + 1 cap (fixed)	54.8	82.2	90.2	35.1	65.6	77.5	56.8	84.0	91.5	49.8	79.9	89.4
BLIP + multi caps	**65.4**	**87.9**	93.7	**43.3**	**73.5**	**83.3**	58.2	85.1	91.8	56.8	84.5	91.5
BLIP + paras	53.1	80.8	88.8	35.9	64.4	75.2	**90.6**	**99.1**	**99.6**	**85.0**	**97.8**	**99.5**
BLIP + caps+paras	60.7	87.5	**94.0**	41.9	72.1	82.6	90.5	98.8	99.5	82.0	97.0	98.9

Table 2. Cross-modal retrieval performance on MS-COCO validation for captions and generated paragraphs. We report Recall@K (%, higher is better) for image-to-text (I2T) and text-to-image (T2I). Best in **bold** and second best underlined.

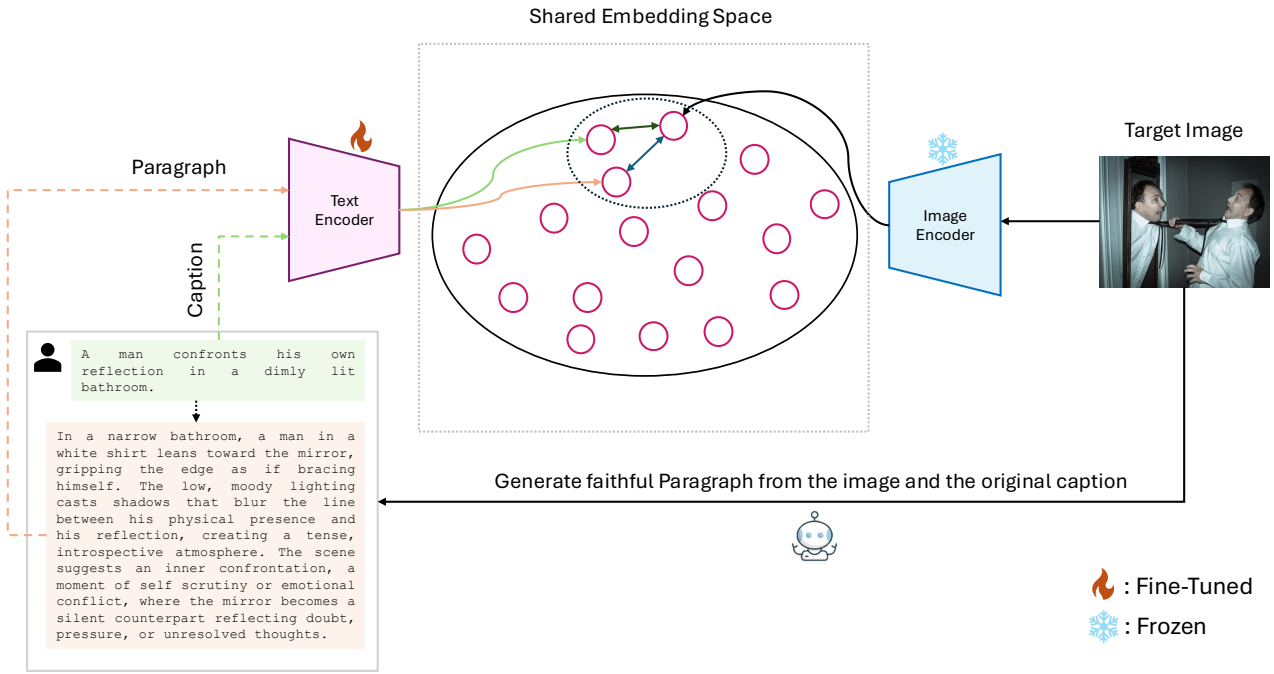

Figure 2. Joint caption plus paragraph training. Each image is associated with multiple short captions and one long paragraph. The multi positive loss encourages the image embedding to be close to all of its descriptions, while pushing it away from descriptions of other images. Only one caption is shown in the figure.

4. Conclusion

We introduced the first large-scale, automatically generated long-form multimodal dataset for MS-COCO, using Qwen2-VL for paragraph synthesis and Llama-3.2 Vision for strict accuracy scoring. Coupled with SPICE validation, this pipeline produces high-quality, globally coherent paragraphs at scale.

We demonstrate that paragraph supervision significantly improves long-form retrieval performance and that combining short- and long-form supervision yields the most semantically aligned contrastive embeddings. Our dataset and training pipeline open a new direction for multimodal pretraining that integrates fine-grained and narrative-level understanding.

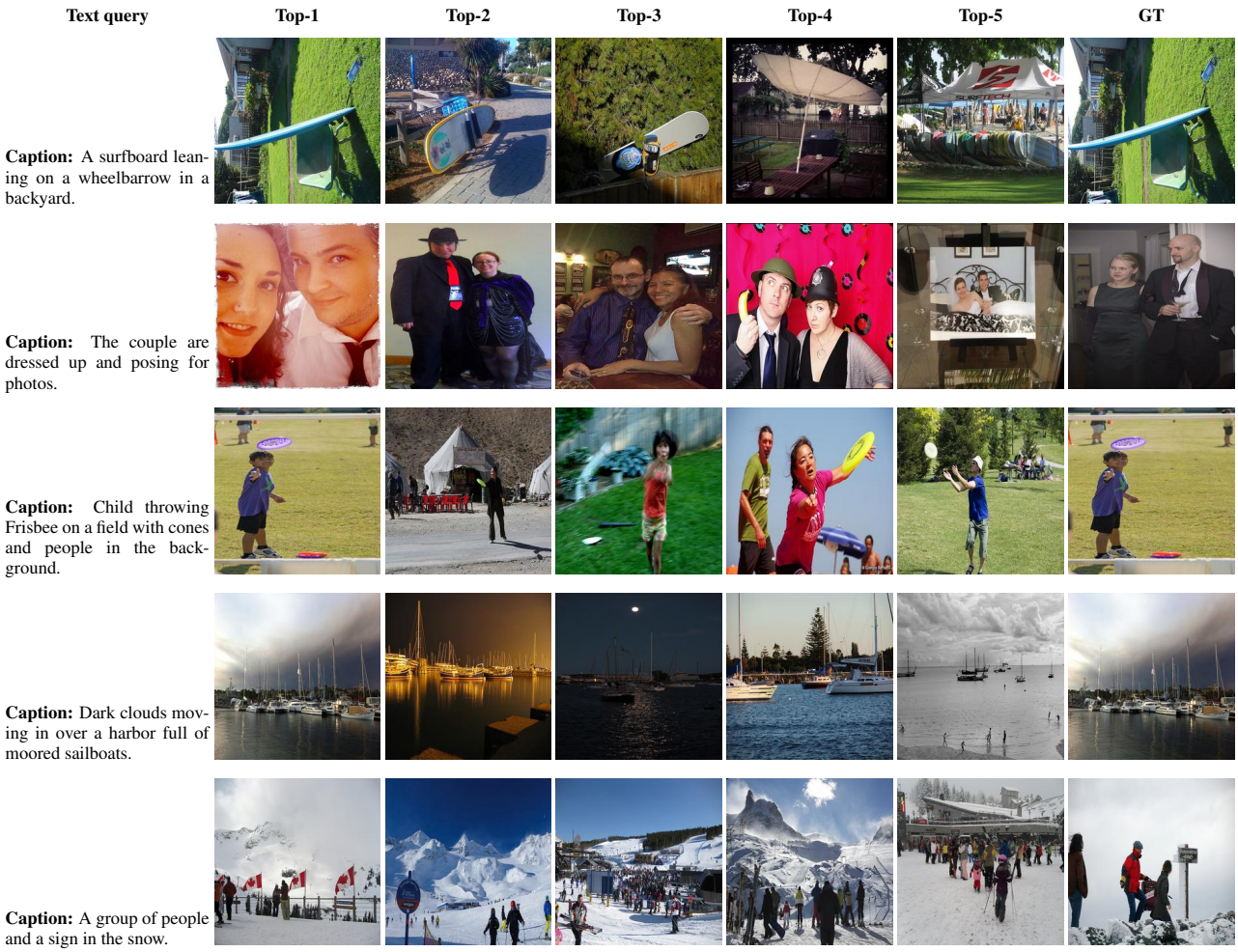

Figure 3. Qualitative caption based retrieval on MS COCO. Each row corresponds to one caption query (shown in the leftmost column). The rightmost image is the ground truth (**GT**), preceded by the top five retrieved images from the joint caption plus paragraph model.

Paragraph Query	Top-1	Top-2	Top-3	Top-4	Top-5	GT
Paragraph: In a vibrant backyard, a blue and white surfboard leans casually against a green wheelbarrow, both resting on a well-maintained lawn. The scene is set against a backdrop of a house with a sloped roof and a garden filled with colorful flowers, creating a picturesque and serene atmosphere.						
Paragraph: The image depicts a couple dressed in formal attire, standing next to each other in a room. The man is holding a martini glass, while the woman is wearing an elegant gray dress. The couple appears to be posing for a photo, possibly at a social event or gathering.						
Paragraph: A young boy stands in a grassy field wearing a purple shirt and black shorts. He has just thrown a purple frisbee, which is seen mid-air. The field is marked with orange cones, and several people are visible in the background, suggesting a recreational or sporting activity.						
Paragraph: A harbor is filled with numerous sailboats docked along the pier. Dark storm clouds gather overhead, casting a moody atmosphere across the calm water, which reflects the boats and sky. The orderly arrangement of the boats evokes a sense of stillness before an approaching storm.						
Paragraph: A group of people walk up a snowy mountain trail, carrying luggage while the wind blows across the landscape. A signpost stands nearby, and the overcast sky emphasizes the cold, harsh conditions of their journey.						

Figure 4. Qualitative **paragraph-based retrieval** on MS COCO. Each row corresponds to a paragraph query (shown in the leftmost column). The rightmost column shows the ground truth image (**GT**), preceded by the **Top-1 to Top-5** retrieved images from the joint caption+paragraph model.